Practice Problems
Calculus-Based Physics II

Name: Key **Date:** _____

Chapter 34 – Electric Charge

1) The fundamental force underlying all chemical reactions is

A) gravitational.
B) nuclear.
C) electrical.
D) centripetal.
E) None of the above choices are correct.

2) In an electrically neutral atom, the number of protons in the nucleus is equal to the number of

A) electrons that surround the nucleus.
B) neutrons in the nucleus.
C) Choices A and B are both correct.
D) Choices A and B are both incorrect.

3) A positive ion has more

A) protons than neutrons.
B) protons than electrons.
C) electrons than protons.
D) electrons than neutrons.
E) neutrons than protons.

4) Strip electrons from an atom and the atom becomes a

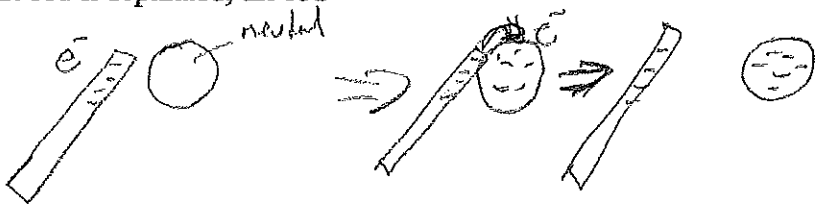
A) different element.
B) positive ion.
C) molecule.
D) negative ion.

5) To say that electric charge is conserved is to say that electric charge

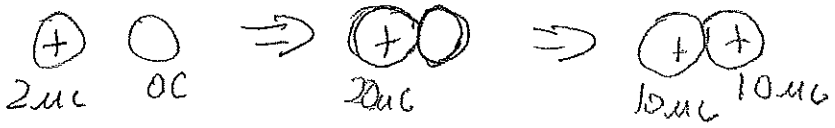
A) will interact with neighboring electric charges.
B) is a whole number multiple of the charge of one electron.
C) is sometimes negative.
D) can be neither created nor destroyed.
E) may occur in an infinite variety of quantities.

6) A plastic rod is charged up by rubbing a wool cloth and brought to an initially neutral metallic sphere that is insulated from ground. It is allowed to touch the sphere for a few seconds, and then is separated from the sphere by a small distance. After the rod is separated, the rod

A) is repelled by the sphere.
B) is attracted to the sphere.
C) feels no force due to the sphere.

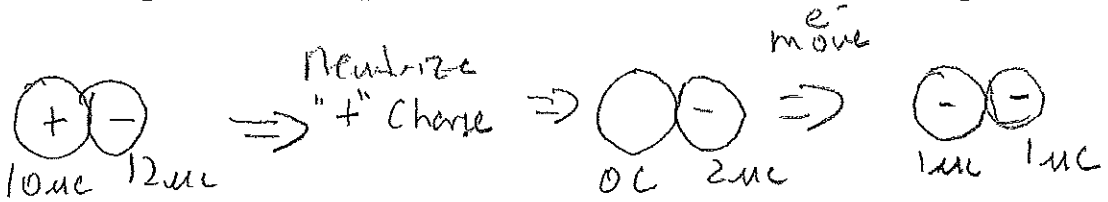


7) Given two identical conducting spheres, Sphere 1 has 20 μC of positive charge and Sphere 2 is neutral, the spheres are brought together, and they touch, the charge equalizes then the sphere are separated, what is the charge on each sphere after they are separated?



10 μC positive

8) Given two identical conducting spheres with different net charge, Sphere 1 has 10 μC of positive charge and Sphere 2 has 12 μC of negative charge, the spheres are brought together, and they touch, the charge equalizes then the sphere are separated, what is the charge on each sphere after they are separated?



1 μC negative

9) The charge carried by one electron is $e = 1.6 \times 10^{-19}$ C and is negative. The number of electrons necessary to produce a charge of 2 μC is

$Q = N q_{e^-}$ $\quad N = \frac{Q}{q_{e^-}} = \frac{2 \times 10^{-6} C}{1.6 \times 10^{-19} C} = 1.25 \times 10^{13}\ e^-$

10) If a charge generator builds a negative static charge of 7 μC, how many electrons are transferred to it during this process. ($e = 1.60 \times 10^{-19}$ C)

$N = \frac{Q}{q_{e^-}} = \frac{7 \times 10^{-6} C}{1.6 \times 10^{-19} C} = 4.37 \times 10^{13}\ e^-$

11) A piece of plastic has a net charge of 2 μC and is positive, how many more protons than electrons does this piece of plastic have?

$N = \frac{Q}{q_p} = \frac{2 \times 10^{-6} C}{1.6 \times 10^{-19} C} = 1.25 \times 10^{13}\ p^+$

12) To say that an object is electrically polarized is to say

A) its charges have been rearranged.
B) its internal electric field is zero.
C) it is electrically charged.
D) it is only partially conducting.
E) it is to some degree magnetic.

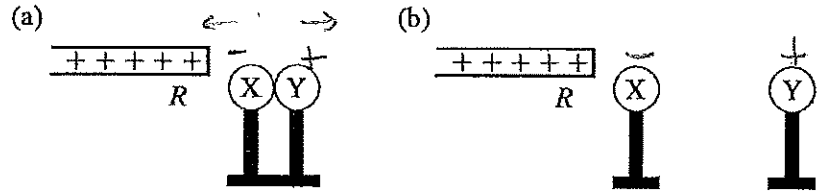
13) A balloon will stick to a neutral wall if the balloon is charged

A) positively.
B) negatively.
C) either positively or negatively.
D) None of the above choices are correct.



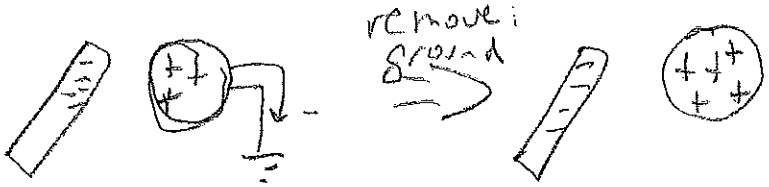
14) X and Y are two initially uncharged metal spheres on insulating stands, and they are in contact with each other. A positively charged rod *R* is brought close to X as shown in part (a) of the figure. Sphere Y is now moved away from X, as shown in part (b). What are the final charge states of X and Y?

A) Both X and Y are neutral.
B) X is positive and Y is neutral.
C) X is neutral and Y is positive.
D) X is negative and Y is positive.
E) Both X and Y are negative.



15) A negatively charged plastic rod is brought close to (but does not touch) a neutral metal sphere that is connected to ground. After waiting a few seconds, the ground connection is removed (without touching the sphere), and after that the rod is also removed. The sphere is now

A) negatively charged.
B) positively charged.
C) neutral.



Answers

1) C
2) A
3) B
4) B
5) D
6) A
7) 10 μC
8) 1 μC
9) 1.25×10^{13} electrons
10) 4.37×10^{13} electrons
11) 1.25×10^{13} protons
12) A
13) C
14) D
15) B